SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

FIN207

Timed Online Assignment - January Semester 2023

Game Theory and Design Thinking

Tuesday, 16 May 2023 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **5** questions and comprises **6** pages (including cover page).

2. You must answer **5** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. **Images of handwritten answers will not be marked**.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

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7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

(Full marks: 100)

Section A (100 marks)
Answer all questions in this section.

Question 1

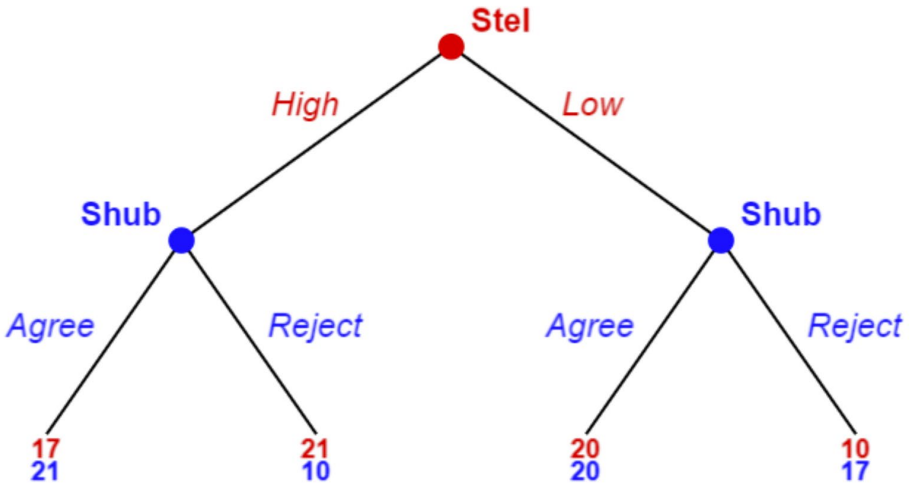
"It is possible to have 3 pure-strategy Nash equilibria in a two-player simultaneous game with two strategies for each player." Judge the statement as being true or false. If it is false, explain. If it is true, fill in the following payoff matrix to show an example.

		Player 2	
		c	d
Player 1	a		
	b		

(10 marks)

Question 2

Consider a 5G technology service pricing game between two telecommunication firms Stel and Shub. Stel is deciding whether to price the new service at a high or low price (Choice: High or Low). Shub is deciding whether to agree or to reject the same pricing strategy (Choice: Agree or Reject). The game tree is as follows:



Question 2a

Interpret the game tree by using a game table and highlight the best responses in the matrix. Note: The row player is Stel, and the column player is Shub.

(16 marks)

Question 2b

Identify the subgame perfect equilibrium in the strategic form from (a). Justify your answer.

(8 marks)

Question 3

Consider the following simultaneous game between Jimmy and Kate:

		Kate		
		H	M	L
Jimmy	H	25,75	80,20	90,10
	M	90,10	20,80	95,5
	L	15,85	10,90	30,70

Question 3a

Indicate if there is any dominant strategy or dominated strategy.

(6 marks)

Question 3b

Show the new payoff matrix after successive elimination of dominated strategies. Provide an example (or scenario) to illustrate this game in real life according to the new payoff matrix.

(15 marks)

Question 3c

Analyse and find all the equilibria (including pure strategy and mixed-strategy equilibria) according to the new payoff matrix from (b).

(10 marks)

Question 4

Consider a game between 2 Hainanese Chicken Rice restaurants. They will decide whether to provide a discount of the price or not next month. The payoff matrix (Unit: Thousand) is as follows:

(Note: "Yes" is to provide the discount, "No" is to not provide)

		Player 2	
		Yes	No
Player 1	Yes	8,4	5,5
	No	6,6	4,8

Question 4a

Is the game a prisoners' dilemma? Judge and explain.

(10 marks)

Question 4b

Without any calculation, judge whether it is possible to have a mixed strategy equilibrium with a probability between 0 and 1 (**excluding 0 and 1**) for this payoff matrix. Explain your judgment.

(5 marks)

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Question 5

Metaverse Company produces high-resolution VR headsets and low-resolution VR headsets. The costs are $3,000 per unit for the high-resolution product and $1,000 per unit for the low-resolution one. Assume that there are two types of buyers, i.e., ordinary and special buyers, in the market. According to a market survey, ordinary buyers are willing to buy the high-resolution headset at $5,000 and low-resolution headset at $4,000. Special buyers are willing to buy the high-resolution headset at $8,000 and low-resolution headset at $5,000.

Question 5a

Assume that Metaverse Company is omniscient, i.e. it can recognize the types of the buyers. Assume that the company sells the products in order to maximize its expected profit according to the cost and the willingness of the buyers. Calculate the profit of selling one unit of product to different types of the buyers and illustrate the optimal selling strategy (i.e. which product should be sold to which type of the buyers).

(10 marks)

Question 5b

Assume that the company cannot recognize the types of the buyers. But it is known that 80% of all buyers are ordinary buyers and 20% are special buyers. If the company decides to produce high-resolution VR headsets only in the future, which price (i.e. 5000 or 8000) should the company set for the high-resolution headsets to maximize its expected profit?

(6 marks)

Question 5c

Assume that the company cannot recognize the types of the buyers and continue to produce both headsets. The company sets the prices of high-resolution and low-resolution VR headsets as pH and pL respectively. If the company wants to motivate the ordinary buyers to choose low-resolution headsets and motivate the special buyers to choose high-resolution headsets, show the incentive-compatibility constraints and participation constraints by using pH and pL.

(4 marks)

----- END OF PAPER -----

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